Progress of The New Spain Fleets Project: accurate Handwritten Text Recognition models for 16th-17th century Spanish calligraphies.

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Introduction

The New Spain Fleets project is an ongoing large research endeavour by an interdisciplinary team of archaeologists, historians and computer scientists to delve into the socioeconomic history of the former Spanish empire's maritime trade. Leveraging Artificial Intelligence methods, we analyse a large historical corpus created in Spain and Mexico between the 16th and 17th centuries.

A major challenge for accessing the information in these documents lies in their calligraphy, terminology, and conventions, which require years of specialised training for reading and analysis. This has limited historical research to relatively few scholars, maintaining most of our written heritage still "locked away" in archives.

Historical documents from the Spanish colonial era comprise a distinctive set of calligraphies, the most prominent being cortesana, redonda, procesal simple, itálica cursiva, procesal encadenada, and pseudorredonda. These types are found throughout the Spanish colonial archive, often presenting regional variations, particularly in documents created in Mexico (Murrieta-Flores et al., 2025).

Handwritten Text Recognition (HTR) models have been successfully developed for the automatic transcription of historical manuscripts, leveraging Convolutional Neural Networks (Nockels et al., 2022; Ferro et al., 2023) although most freely available models have been designed for Northern European languages, and few have only recently become available for Spanish (Ball et al., 2021; Cuéllar, 2023b; Cossío Olavide, 2023; Cuéllar, 2023a). However, in our experience, the latter still do not yield accurate transcriptions, either because of being trained for specific sets of documents or too heterogeneous datasets with documents spanning several calligraphies and centuries. Large Spanish models available perform best with late (19th–20th century) documents, their accuracy greatly diminishing when working with earlier documents and "dirty" calligraphies such as procesal simple. To tackle these challenges, we trained specific models, carefully classifying the documents by calligraphy and generating a large sample of human transcription by specialist palaeographers.

Here, we describe advances and results in developing four accurate HTR models for the automatic transcription of 16th–17th-century Spanish calligraphies: Itálica cursiva, Procesal simple, Redonda, and Procesal encadenada.

Methods

We trained both HTR models using Transkribus. The workflow involved 1) the creation of a 170-document corpus classified by calligraphy; 2) uploading the digitised images of the historical documents, obtained from the archives' websites or manual digitisation; 3) preparing the Ground Truth datasets; 4) training the models; 5) iterating until a Character Error Rate (CER) of <10% on the validation sample was obtained; 6) preparing test datasets and performing accuracy tests.

Ground Truth refers to the images and the corresponding accurate transcriptions on which a model will learn during training. All transcriptions were carried out manually by a group of palaeographers specialised in New Spain calligraphies. One team member prepared and another revised/corrected literal transcriptions, maintaining spelling, grammar, and punctuation as appeared in the originals, without expanding abbreviations.

Historical documents in the Ground Truth datasets (Table 1), currently held in major public archives from Mexico, France, Spain, and the USA, were originally produced in New Spain (today's Mexico) or Spain between the 16th and 17th centuries. All were written in Spanish, although some contain words (e.g., toponyms) in their original indigenous language, mainly Nahuatl. We only included words in Spanish in the training datasets.

Table 1. Ground Truth datasets.

	Itálica cursiva (m3t1)	*Procesal simple (m3t4)*	Redonda (m1t3)	*Procesal encadenada (m2t2)*
Documents	40	76	12	4
Number of pages	339	1,148	319	410
Number of lines	10,579	38,725	1,804	15,237
Number of words	106,046	258,503	21,860	111,807

One distinctive aspect of our workflow was adjusting line polygons to ensure they completely encapsulated the letters. Although it has been discussed elsewhere that this process might not always be necessary, this was particularly crucial for the *Procesal simple* calligraphy in which letters usually exceed line limits (Figure 1). We used the adjusted line polygons for training, but all other basic and advanced training settings were set to default in Transkribus (i.e., epochs=100, learning rate=0.0003, batch size=24). The Validation set comprised 10% of the Ground Truth sample.

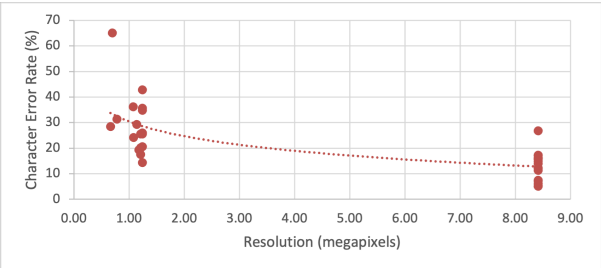

Figure 1. Adjusting line polygons.

We performed accuracy tests for each HTR model using the Transkribus Compute Accuracy tool (Transkribus Expert Client v1.28.0), using documents not included in training or validation. The testing datasets included 60 pages (years 1554 to 1712) for the *Itálica cursiva* model, 35 pages (1590-1608) for the *Procesal simple* model, 21 pages (1570-1580) for the *Redonda* model, and 13 pages (1621) for the *Procesal encadenada* model.

Results

The *Itálica cursiva* model was trained on 97 documents from 1536 to 1717. Its first three iterations comprised relatively small datasets (<82,000 words), not achieving the expected CER. The fourth iteration (Itálica_cursiva_m3t1) included over 106,000 words, achieving a CER=8.8% without a base model. Accuracy tests yielded a median CER=7.61% (IQR=13.23%) on the test sample. Text comparisons showed that most automatic transcriptions were accurate, and errors would be negligible to a trained reader (*Figure 2*). Furthermore, a comparison with the largest freely available HTR model for Spanish (Coloso Español, CER=4.8%) showed that our model significantly outperforms the former (CER=23.70% ColEsp vs 15.97% ItalCur_m3t1).

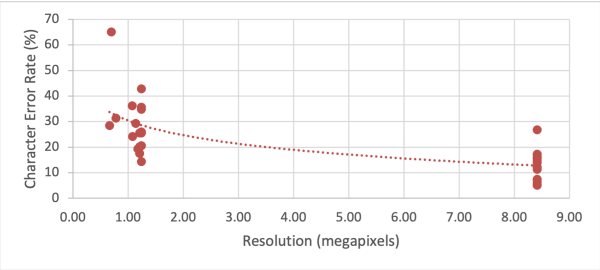

Figure 2. Text comparison for an Italica cursiva HTR transcription.
The test only highlights those words where Ground Truth (green) and HTR (red) differ.

The *Procesal simple* HTR model was trained on over 80 documents from 1512 to 1657, involving nine iterations. The last iteration (Procesal_simple_m3t4) included an accumulated sample of 258,503 words, achieving a CER=8.92%. Accuracy tests showed a median CER=16.72% (IQR=12.23%). This higher CER was affected by two factors. First, image resolution: low-resolution images from an archive's website (≤1.24 MP, n=17) yielded significantly higher CERs than manually digitised images (8.41 MP, n=18) (median=26.04% vs 14.53%, p<0.001, Figure 3). The negative effect of image resolution on model recognition accuracy has long been reported and continues to be an important challenge in character recognition (Jacobs et al., 2005; Baek et al., 2019; Wang et al., 2020). Second, calligraphy "neatness": A subset of documents with clearer calligraphy yielded even lower CERs than those in the validation sample (median=6.1%, IQR=2.04%).

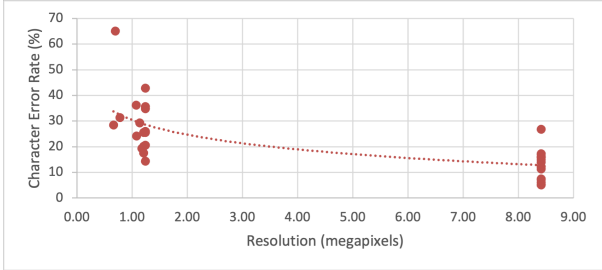

Figure 3. Impact of image resolution on the *Procesal simple* model accuracy.

The *Redonda* and *Procesal Encadenada* models are our latest developments. The *Redonda* model was trained on twelve documents from 1536 to 1576. Although it was a relatively small dataset, the homogeneity and clarity of this calligraphy style allowed the model to achieve a CER=5.25% when trained with the *Procesal Simple* as a base model.

The *Procesal Encadenada* model recognises a particularly challenging calligraphy. It required the preparation of a large training set by a group of expert palaeographers, comprising over 111,000 words in documents from 1590 to 1640. Leveraging the *Procesal Simple* as base model, the model achieved a CER=7.80% on the validation set. Accuracy tests showed a median CER=18.17% (IQR=6.23%) on the testing set while also confirming its outperformance of the largest freely available model, Coloso Español, on this sample (CER=33.55%, IQR=6.94).

Conclusions

We developed HTR models for *Itálica cursiva, Procesal simple, Redonda,* and *Procesal encadenada* calligraphies (CER=8.8%, 8.92%, 5.25%, and 7.80%, respectively), four of the most prevalent in 16 th-17 th-century Spanish historical documents. The models provide sufficiently accurate transcriptions, allowing users to read the text fluidly, requiring minor corrections. Considering the significant effect of image quality on models' accuracy, users and cultural heritage institutions should work with high-resolution images whenever possible. Our models will soon become publicly available as outputs of the New Spain Fleets project, allowing interested audiences to access vast amounts of historical information so far locked behind challenging Spanish American calligraphies.

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